```sql
CREATE TABLE zepto (
    sku_id INT IDENTITY(1,1) PRIMARY KEY,
    category VARCHAR(120),
    name VARCHAR(150) NOT NULL,
    mrp NUMERIC(8,2),
    discountPercent NUMERIC(5,2),
    availableQuantity INT,
    discountedSellingPrice NUMERIC(8,2),
    weightInGrm INT,
    outofStock BIT,
    quantity INT
);





ALTER TABLE zepto_pro
ALTER COLUMN outofStock VARCHAR(10);



ALTER TABLE zepto_pro
ADD sku_id INT IDENTITY(1,1) PRIMARY KEY;

select * from Zepto_pro


--#exploring

select count(*) from zepto_pro

select top 10 * from zepto_pro

--null value

select * from zepto_pro
where name is null

OR
Category is null
or
mrp is null
or
discountPercent is null
or
discountedSellingPrice is null
or
```

```sql
availableQuantity is null
or
weightInGms is null

or
outofStock is null
or
 quantity is null

--change the column value 0 =no and 1 =yes

UPDATE zepto_pro
SET outofStock =
    CASE
        WHEN outofStock = '1' THEN 'Yes'
        WHEN outofStock = '0' THEN 'No'
    END;

 -- different product categories
 select distinct category
 from Zepto_Pro
 order by category

 --- products in stack vs out of stock
 select outofStock , count(sku_id)
 from zepto_pro
 group by outofStock

--products names present multiple times

select name, count(sku_id) as 'Number of SKUs'
from zepto_pro
group by name
having count(sku_id)>1
order by count(sku_id)desc


--data cleaning

---products with price =0

select * from Zepto_Pro
where mrp = 0 OR discountedSellingPrice=0


delete from Zepto_Pro
where mrp=0

--covert paise to rupess
```

```sql

update Zepto_Pro
set mrp=mrp/100.0,
discountedSellingPrice=discountedSellingPrice/100.0

select mrp, discountedSellingPrice from Zepto_Pro
--

--1. Find the top 10 best value products based on the discount percentage
select DISTINCT
 TOP 10
name,
mrp,
discountPercent
from zepto_pro
order by discountPercent desc;

--02 What are the Products with High MRP but Out Of Stack
SELECT DISTINCT
    name,
    mrp
FROM zepto_pro
WHERE outOfStock = 'Yes'   -- ☑ correct
  AND mrp > 300
ORDER BY mrp DESC;

--03 Calculate Estimated Revenue for each category
select category ,
sum(discountedSellingPrice * availableQuantity) AS total_revenue
from zepto_pro
group by category
order by total_revenue

--04 Find all products where mrp is greater than  rs500  and discount is less than 10%
SELECT DISTINCT name,mrp,discountPercent
FROM  Zepto_Pro
WHERE mrp > 500 AND discountPercent<10
ORDER BY mrp,discountPercent DESC;

--05 Identify the top 5 category offering the highest average discount percentage

select TOP 10 category,
ROUND(AVG(discountPercent),2) AS avg_discount
from Zepto_Pro
group by category
order by avg_discount desc;

```

```sql

--06 find the price per gram for products above 100gm and sort by best value
select distinct name,
weightInGms,
discountedSellingPrice,
ROUND(discountedSellingPrice/weightInGms,2) AS price_per_gram
from zepto_pro
where weightingms>=100
order by price_per_gram ;


--07 Group the products into categories Like low, Medium ,Bulk

select DISTINCT name,weightingms,
case when weightinGms <1000 THEN 'LOW'
when weightingms <5000 THEN 'Medium'
else 'bulk'
end as weight_category
from zepto_pro


--08 what is the total Inventory Weight Per Category


SELECT
--(BIGINT can store very large numbers)--Always use CAST(... AS BIGINT)
    category,
    SUM(CAST(weightInGms AS BIGINT) * CAST(availableQuantity AS BIGINT)) AS total_weight
FROM zepto_pro
GROUP BY category
ORDER BY total_weight DESC;
```